

Big Data Computing

**Written test 21/06/2022
(SOLUTIONS of PART2)**

EX-1 Variant A. Let P be a dataset containing N outcomes of slot machines, represented by the key-value pairs $(i, (s_i, o_i))$, for $0 \leq i < N$, where s_i is a slot machine, and o_i is an outcome of the machine. An outcome o is called *frequent for a slot machine s* if P contains $\geq N/50$ pairs with value (s, o) , and a slot machine s is called *biased* if there is at least a frequent outcome o for s .

1. Design a 3-round MapReduce algorithm that for every biased slot machine s returns exactly one pair (s, null) . (**HINT:** in the first 2 rounds, compute the pairs (s, o) where o is frequent for s .)
2. Analyze the local and aggregate space required by your algorithm.

For full score, your algorithm must require $o(N)$ local space and $O(N)$ aggregate space.

Solution.

1. The 3-round MapReduce algorithm is the following.

Round 1:

- *Map phase:* map each pair $(i, (s_i, o_i))$, into the intermediate pair $(i \bmod \sqrt{N}, (s_i, o_i))$.
- *Reduce phase:* for each $0 \leq j < \sqrt{N}$ separately, let L_j be the list of values (s, o) from the intermediate pairs with key j , and for each pair (s, o) occurring in L_j return the pair $((s, o), k_{s,o}(j))$, where $k_{s,o}(j)$ is the number of occurrences of (s, o) in L_j . (Note that (s, o) is the key of the pair.)

Round 2:

- *Map phase:* empty.
- *Reduce phase:* for each key (s, o) separately, let $L_{s,o}$ be the list of values $k_{s,o}(j)$, with $j \in [0, \sqrt{N})$, from the pairs with key (s, o) produced by Round 1, and if the sum of values in $L_{s,o}$ is $\geq N/50$, then return (s, o) otherwise return nothing.

Round 3:

- *Map phase:* empty.
- *Reduce phase:* for each machine s separately, let L_s be the list of outcomes o from the pairs (s, o) produced by Round 2 (i.e., L_s is the list of frequent outcomes for s), and return (s, null) .

2. In the map phase of Round 1, each application of the map function requires constant space, and in the reduce phases of Round 1 and 2, each reducer processes at most $\sqrt{N}$ elements. As for Round 3, observe that for any each slot machine s there exist at most $50 = O(1)$ frequent outcomes. Based on this observation, in the reduce

phase of Round 3 each reducer processes at most 50 elements, hence it contributes a constant term to the local space. Therefore, the local space is $M_L = O(\sqrt{N})$. The aggregate space is $O(N)$ since input, output and intermediate pairs in each round are $O(N)$.

□

EX-1 Variant B. Let P be a dataset of an online store containing N key-value pairs $(i, (c_i, p_i))$, for $0 \leq i < N$, where c_i represents a customer, and p_i represents a product purchased by c_i in a visit to the store. A product p is *frequent for a customer* c if P contains $\geq N/50$ pairs with value (c, p) , and a customer c is called *compulsive* if there is at least a frequent product p for c .

1. Design a 3-round MapReduce algorithm that for every compulsive customer c returns exactly one pair (c, null) . (**HINT:** in the first 2 rounds, compute the pairs (c, p) where p is frequent for c .)
2. Analyze the local and aggregate space required by your algorithm.

For full score, your algorithm must require $o(N)$ local space and $O(N)$ aggregate space.

Solution. The solution is identical to the one of Variant-A with the following changes of terminology:

- slot machine $\rightarrow$ customer;
- outcome $\rightarrow$ product;
- biased slot machine $\rightarrow$ compulsive customer.

□

EX-2. Let A_1 and A_2 be two n -bit arrays used as Bloom filters to assess the membership for two sets S_1 and S_2 , respectively, from the same universe U . Suppose that both arrays have been initialized by applying the same set of k hash functions $h_1, h_2, \dots, h_k$, to S_1 (in case of A_1) and S_2 (in case of A_2).

1. Describe how to combine A_1 and A_2 into a new n -bit array A and how to use it to assess membership for $S_1 \cup S_2$, with no false negatives (i.e., elements of $S_1 \cup S_2$ erroneously claimed not to be in $S_1 \cup S_2$).
2. Give an estimate of the probability that $A[i] = 0$, for any $i \in [0, n - 1]$.

Solution.

1. For every $0 \leq i < n$, set $A[i] = A_1[i] \text{ OR } A_2[i]$. Let $S = S_1 \cup S_2$. For any element $y \in U$, we claim y to be in S if and only if $A[h_j(y)] = 1$, for $1 \leq j \leq k$. Consider an arbitrary element $x \in S$ and suppose, without loss of generality, that $x \in S_1$, which implies that $A_1[h_j(x)] = 1$, for $1 \leq j \leq k$. From the above definition of A we immediately conclude that $A[h_j(x)] = 1$, for $1 \leq j \leq k$, hence x will be correctly claimed to be in S .
2. Let $m = |S|$. We have that $A[i] = 0$ if and only if for each $x \in S$ and for each $1 \leq j \leq k$, we have that $h_j(x) \neq i$, which happens with probability $(1 - 1/n)^{km} \simeq e^{-km/n}$.

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